JUNE YONG LEE

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PYTORCH NUMPY DOCKER REACT PYTHON NODE.JS TENSORFLOW PANDAS MYSQL REDUX SELENIUM JAVA

Hello, I am June-yong Lee, currently pursuing a master's degree in computer engineering at Hankuk University of Foreign Studies. I am interested in applying diffusion models to autonomous driving research, with a particular focus on deep learning computer vision, generative AI, and multimodal AI. One of my research papers, titled "Lightweight Temporal Segment Network for Video Scene Understanding: Validation of Driver Assault Detection," was published in the Journal of the Korean Institute of Information Scientists and Engineers (KIISE). I also have experience developing and deploying responsive web interfaces for web crawling servers, making me proficient in frontend, backend, and web crawling technologies. My current research interests include transformer-based diffusion models.

Research Experience

1st semester 2024

Lightweight Temporal Segment Network for Video Scene Understanding: Validation in Driver Assault Detection

Korean Institute of Information Scientists and Engineers - Journal of KIISE(JOK)

There has been increasing number of driver assaults in transportation such as taxis and buses in the past years. It can be especially difficult to respond quickly to assaults on drivers by drunks late at night. To address this issue, our research team proposes a lightweight CNN-based Temporal Segment Network (TSN) model, which can detect driver assaults by passengers in real time. The TSN model efficiently processes videos by sampling a small number of image frames and is divided into two streams for learning: one for spatial information processing and the other for temporal information processing. Convolutional neural networks are employed in each stream. In this research, we apply a lightweight CNN architecture, MobileOne, significantly reducing the model size while demonstrating improved accuracy even with limited computing resources. The model is expected to contribute to rapid response and prevention of hazardous situations for drivers by being integrated into vehicular driver monitoring systems.

https://github.com/HUFSuperman/HUFS_SavingDriver

Temporal Segment Network | Lightweight CNN Architecture | Assault Recognition | Abnormal Behavior

2nd semester 2023

Research on Developing a Responsive Web Interface for Small Businesses Using React

Although large corporations have established sales databases for efficient sales activities, small-scale entrepreneurs need a system that provides an interface that anyone can easily manage, offering diverse and complex information. This paper proposes the use of React technology, which can also operate on crawling servers that automatically collect store information. Furthermore, it suggests the development of a responsive web user interface that can be conveniently used in both mouse and keyboard-based desktop environments as well as touch-based smartphone environments, utilizing React technology for the effective delivery of various data. The proposed methods anticipate rapid retrieval of large volumes of data due to the adoption of asynchronous function processing in React technology, an improvement over traditional JavaScript methods.

https://github.com/yousirong/CAPSTONE_AWS

Responsive Web Interface | React | Asynchronous Processing | Information Architecture Design

Education

Korea University of Foreign Studies, Graduate School of Computer Science, Master course [2024~]

Korea University of Foreign Studies, Computer and Electronic Systems Engineering Major [Transfer][2021~2023]

University of Anyang, Computer Science Major [Change of Major][2018]

University of Anyang, Statistics and Data Science Major [Admission][2017]

Awards

Capstone Design Project Excellence Award

Achieved outstanding performance in the Capstone Design Project organized by the AI Education Institute at the Korea University of Foreign Studies.

팀명 : 각자의새벽

Lg Aimers 5th class

Ranked 43rd out of 740 teams in the product defect determination project.

팀명 : 각자의새벽

Research Projects

1. Conversion of Restored Rrs Data to Chl-a and Generation of Difference Map

Objective:

Convert restored Remote Sensing Reflectance (Rrs) data into Chlorophyll-a (Chl-a) and visualize spatial distribution errors through a Difference Map using composite data.

Main Tasks:

- Process GOCI data captured 8 times a day into a single daily dataset by averaging pixel values.
- Select specific coordinates in the Saemangeum or Nakdong River region to convert restored Rrs data to Chl-a and analyze results.
- Calculate the difference between restored results and actual GOCI data, visualizing it using MinMaxScaler with a range of -20 to +20, represented in color scale.

Outcome:

While successful synthesis results from Rrs to Chl-a are rare globally, the study introduced a quantitative evaluation method using Difference Maps to assess restoration performance.

2. Development of a Supervised Learning Model for Restoring Gaps in Chl-a Composite Data

Objective:

Develop a deep learning model to restore missing regions in Chl-a composite data, averaged over 8 days.

Main Tasks:

- Convert UST21 dataset into 8-day moving averages for model training. For example, calculate averages for January 1–8 and January 2–9, and so on.
- Focus on central regions around Saemangeum and Nakdong River but adjust the area towards the open sea if data availability is limited.
- Train the RFRNet model with masked data for restoration tasks.

Validation:

- Compare restored results with MODIS 8-day average data to evaluate performance.
- Conduct quantitative analysis of restoration results for Nakdong River and Saemangeum regions using RMSE, MAE, and R^2 metrics.
- Apply smoothing techniques for missing data when testing images larger than the traditional 256x256 patch size.

Technologies and Tools

- **Data Processing:** Python, NumPy, PyTorch, TensorFlow
- **Satellite Data:** GOCI Rrs, UST21 Chl-a, MODIS Chlorophyll-a
- **Deep Learning Model:** RFRNet (Restoration Focused Refinement Network)
- **Data Visualization and Analysis:** Matplotlib, Seaborn, RMSE/MAE evaluation metrics

Achievements

- Proposed a new evaluation metric using Difference Maps based on restored Rrs and Chl-a data.
- Enhanced satellite image restoration technology by leveraging UST21 and MODIS datasets.
- Optimized restoration of missing data, increasing practical applicability of research outcomes.

Research Contributions

Improved the utility of ocean satellite data by analyzing restoration and prediction model performance, ensuring academic and industrial applicability.